It was never the geometry:

the remapped *dynamics seam* is what kills FESOM on a moving-cavity leg

Findings of 2026-07-13 — supersedes the balance-consistent restart plan

2026-07-13

PARTIALLY REFUTED (2026-07-13, evening). The core diagnosis in this note stands: the cavity *geometry* is fine, a cold start on it is stable, and the killer is the remapped *dynamics seam*. **But its central premise about the FIX is wrong.** This note argues the per-leg PISM increment is "two orders of magnitude smaller" (mean $|\Delta \text{draft}| \approx 22$ m) and therefore absorbable. It is not. The increment test was run: **1399 nodes have their ice removed ENTIRELY in one leg**, the Ross front retreats ≈440 km, and PISM loses 24% of its floating area per decade. The mean is diluted by the 98% unchanged interior and is **meaningless**. Struck-out passages below are the refuted ones. See `report/increment_test_2026-07-13`.

Abstract

This note reports a set of controlled experiments that **falsify the previous root-cause claim** of this investigation. The moving-cavity blowup is *not* a geostrophically unbalanced restart, *not* the basal-melt flux shock, *not* the timestep, and *not* the cavity geometry. A **cold start on the very same PISM cavity, at the full production timestep, is stable**; the identical mesh warm-started from our remapped restart dies on day 4. What kills the model is the **seam between the evolved dynamics we carry over on the untouched 98.2% of the mesh and the patched values at the changed nodes**. We also show that the change we have been testing all along — the one-off CORE3(observed)→PISM cavity swap — is a pathological monster (max $|\Delta \text{draft}| = 2711$ m) that production never has to survive, whereas the change production *does* have to survive, a 10-yr PISM increment, has **never been tested**. The strategy therefore changes: cold-start the ocean on PISM's cavity and never remap across the swap at all.

1 What is now falsified (including our own previous conclusion)

The design note `balanced_restart_plan.tex` argued the blowup was a missing geostrophic jet and proposed initialising the changed-cell velocity in thermal-wind balance. That was implemented (per-level $\mathbf{u}_g = (1/f\rho_0)\,\hat{z} \times \nabla p$ using FESOM's own cavity pressure convention). It **failed its own offline gate**: the discrete ∇p at grid-scale fronts is *noisier* than the nearest-neighbour fill (transport divergence above the NN baseline), so it ships **opt-in only** (`REMAP_GEOSTROPHIC=1`) and is off by default. It is not the fix, and the hypothesis behind it is dead.

Every hypothesis we have held is now eliminated by a direct experiment:

Hypothesis	Experiment	Verdict
Atmosphere / OASIS / coupling	ocean-only standalone FESOM reproduces the crash exactly (day 4, CFLz 9.24)	**ruled out**
CORE2 forcing shock (standalone artefact)	full-mesh control, *same* forcing + native restart: stable 21 d, worst CFLz **2.61**	**ruled out**
Remap corrupting the state	remapped vs. source over the 204 970 unchanged nodes: max $|\Delta| = \mathbf{0.000}$ for `temp`, `salt`, `ssh`, `hnode`	**ruled out** (bit-exact)

Hypothesis	**Experiment**	**Verdict**
The restart *fill values* at changed nodes	five independent fill strategies (v2–v8: donor, coherent-column, shelf-base hold, NN velocity, full balance hygiene)	**ruled out** (all die ≈day 4)
Pathological thin water columns	submesh min wet column 30 m vs. mother's 25 m; min 3 wet layers on changed nodes; zero 1–2-layer columns	**ruled out** (submesh is *cleaner*)
Basal melt flux shock	`cavity_flux_ramp_days` implemented; melt effectively **OFF** still dies (mstep 328)	**ruled out**
Timestep / unresolved w	1200 s: CFLz kill day 4.0. 60 s: **0** CFLz warnings, dies day 4.4 via η_n. 120 s: dies day 8.9	**ruled out**
Sub-ice free-surface BC violation	real bug found + fixed (987 nodes, below); run still dies day 4.4	**real bug, not the cure**
Cavity geometry itself	**cold start on the same PISM submesh @1200 s: STABLE, day 13+, worst CFLz 3.44**	**ruled out**

2 The decisive control: a cold start on PISM's cavity is stable

The single experiment that inverts the picture. Same submesh, same forcing, same production timestep — only the initial state differs:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up CORE3 restart	stable 21 d, worst CFLz **2.61**
PISM cavity, cold	**1200 s**	**cold (climatology, at rest)**	**stable 13+ d, worst CFLz 3.44**
PISM cavity, warm	1200 s	our remapped restart	**dies day 4.0**, CFLz 9.24
PISM cavity, warm	120 s	our remapped restart	dies day 8.9 (η_n)
PISM cavity, warm	60 s	our remapped restart	dies day 4.4 (η_n, 0 CFLz)

The ocean has no difficulty living in PISM's cavity at the production timestep. Its CFLz (3.44) sits essentially at the full-mesh control's background level (2.61). Whatever is wrong is in *our restart*, not in the mesh, the physics, or the coupling.

3 What is actually wrong: the dynamics seam

The remap produces a state that is internally *inconsistent*:

- on the **204 970 unchanged nodes** (98.2%) it reproduces the evolved restart **bit-exactly** — a fully developed, geostrophically balanced dynamic state ($\mathbf{u}$, η, AB history);
- on the **3840 changed nodes** it substitutes *patched* values (donor-filled tracers, NN or zero velocity, donor η).

Those two regions are mutually incompatible, and the **seam between them** is the instability. The evidence is consistent and specific:

- the CFLz runaway grows *monotonically* (4.46 → 6.43 → 7.25 → 7.51 → 7.92 → 9.24);

- every escalation site lies **within** 0.16–0.22° **of a changed node**, and the worst one *is* a changed cavity node — it grows in the ring around the patch and spreads;
- it is independent of melt (survives melt-OFF) and of Δt (60 s merely swaps the *symptom* from CFLz to η_n);
- **zeroing the dynamics globally** (keep remapped T, S; set $\mathbf{u} = \eta = \bar{h} = w = \mathrm{AB} = 0$, `hnode` nominal) **removes the seam and the crash** — the ocean then re-derives its own balanced flow from rest, exactly as the cold start does.

This finally explains why *five* fill strategies all failed identically: each one varied the *patch* while leaving the *seam* in place. The problem was never the values we put at the changed nodes; it is that we splice *any* patch against a fully evolved dynamical field.

But zero-dynamics is not a production fix. Discarding the ocean's circulation at every coupling leg is unacceptable — the whole point is to remap restarts for an ever-evolving PISM geometry and have FESOM *survive with its dynamics intact.* It is a diagnostic, not a cure.

4 A real bug found along the way (not the cure)

FESOM keeps $\eta \equiv 0$ under an ice shelf: in the source restart, every already-sub-ice node is *exactly* 0.000. The remap, however, carried the old *open-ocean* value across for nodes the mesh change puts **newly under ice**:

node class	count	`ssh`
still-ice (ice before *and* after)	2368	+0.0000 (exactly)
newly-ICE (open before, sub-ice now)	**987**	$-$**1.60** m (all nonzero)
open-both	204 797	$-$0.46 (normal range)

987 nodes entered the ssh solver with a free surface it can never satisfy — a standing violation of the model's own surface BC, with `hbar` corrupted identically. This is a genuine correctness bug and is **fixed** (zero `ssh`/`hbar` at `ulevels_nod2D > 1`). It is **not** the cure: the fixed run still dies on day 4.4. We report it because it was a plausible-looking culprit that explained much of the evidence (melt-independent, Δt-independent, barotropic, margin-localised) and *was still wrong* — a caution for the next hypothesis.

5 We were testing the wrong change

The most consequential realisation. The mesh change we have spent this entire investigation fighting is the **one-off CORE3(observed) → PISM cavity swap**, which is pathological:

max $\lvert\Delta\text{draft}\rvert$ (observed → PISM)	
nodes with $\lvert\Delta\text{draft}\rvert > 200$ m	
nodes flipping open-ocean → sub-ice	
nodes flipping sub-ice → open-ocean	
~~**a genuine 10-yr PISM increment**: mean $\lvert\Delta\text{draft}\rvert$~~	~~**≈22 m**~~ ← **REFUTED: the mean is meaningless; 1**

This swap is an artefact of bolting an ocean spun up on the *observed* cavity onto PISM's *equilibrium* cavity — two different ice sheets. **Production never has to survive it.** ~~What production must survive, forever, is the *incremental* change from one PISM chunk to the next — two orders of magnitude smaller —~~ and **that has never been tested**. **REFUTED:** the increment is *not* "two orders of magnitude smaller". PISM is in disequilibrium and sheds its shelves: 1399 nodes lose their ice outright in a single leg (Ross front −440 km, floating area −24%/decade). The strike-through above is the error.

6 The strategy that follows

1. **Never remap across the swap. Cold-start** FESOM *on PISM's cavity* in chunk 1 (§2 proves this is stable at 1200 s). The observed cavity never enters the coupled run. This is experiment **movcav12** (running).
2. **Then test the change that matters.** By chunk 3 the cold-started ocean carries a year of genuine, spun-up dynamics; the remap must then carry *that* across a real 10-yr PISM increment ~~(≈22 m)~~ (in fact: 1399 nodes fully opened). **This is the experiment that decides whether the coupling is viable**, and it is the first honest test of the remap we will have run.
3. **If the increment survives**, the machinery is fit for purpose — we have already proven the remap is bit-exact on unchanged nodes, inversion-free, and geometrically sound.
4. **If even ~~≈22 m~~ a real increment dies**, the seam is fundamental and the remaining option is to stage the geometry change so the ocean never sees a discontinuity (a geometry ramp), or to relax the dynamics locally toward balance across the seam rather than patching it.

7 Assets built (reusable)

- **A 2-minute offline reproducer**: standalone ocean-only FESOM (`setup_name: fesom` via esm_tools) on the PISM-cavity submesh, 14 nodes, crashes in ∼2 min — versus the 45-node coupled system. Runscript: `scripts_is/sa_spin_movcav11.yaml`. Requires a *standalone* binary (`build_sa`, `FESOM_COUPLED=OFF`) rebuilt at the current SHA; note FESOM puts its code in `libfesom.so` and `fesom.x` is a thin wrapper whose timestamp never changes on rebuild.
- **A stable control**: full mesh + native restart + same forcing (21 d, CFLz 2.61) — the reference every experiment is measured against.
- `cavity_flux_ramp_days` (FESOM `&run_config`, default `0.0` = off, bit-identical): ramps cavity basal heat/freshwater fluxes 0 → 1 over N model days. Melt was exonerated, but the knob is a clean, off-by-default spin-up device.
- Sub-ice `ssh`/`hbar` zeroing in the remap (§4).
- `verify_balance.py`: offline balance/stability gate (min N^2, max $|\nabla\cdot(H\bar{\mathbf{u}})|$, max $|\mathbf{u}|$, CFL, Ri) — this is what caught the geostrophic ∇p init being worse than the NN baseline *before* it cost a model run.

8 Method note

Two conclusions in this investigation were asserted with more confidence than the evidence supported ("geostrophic imbalance is the root cause"; "the sub-ice ssh BC violation is the root cause"). Both were later falsified by a direct control. The controls that actually cracked it — the **full-mesh**

control (is it our setup or the cavity?) and the **cold start** (is it the geometry or our state?) — are both cheap, and both should have been run *before* building fixes on top of an unverified hypothesis.